

ASSIGNMENTS

Class-3

Geometry of Linear Equations

1. Find two points on the line of intersection of three planes $t=0$, $z=0$ and $x+y+z+t=1$
2. Describe the intersection of the three planes $u+v+w+z=6$, $u+w+z=4$, & $u+w=2$. Is it a line or a point or an empty set? What is the intersection of the fourth plane $u=-1$? Find the fourth equation that leaves the system with no solution.
3. Under what condition on y_1, y_2, y_3 do the points $(0, y_1)$, $(1, y_2)$, $(2, y_3)$ lie on a straight line?

